

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technology Stack — Electricity Consumption Analytics (Tableau)

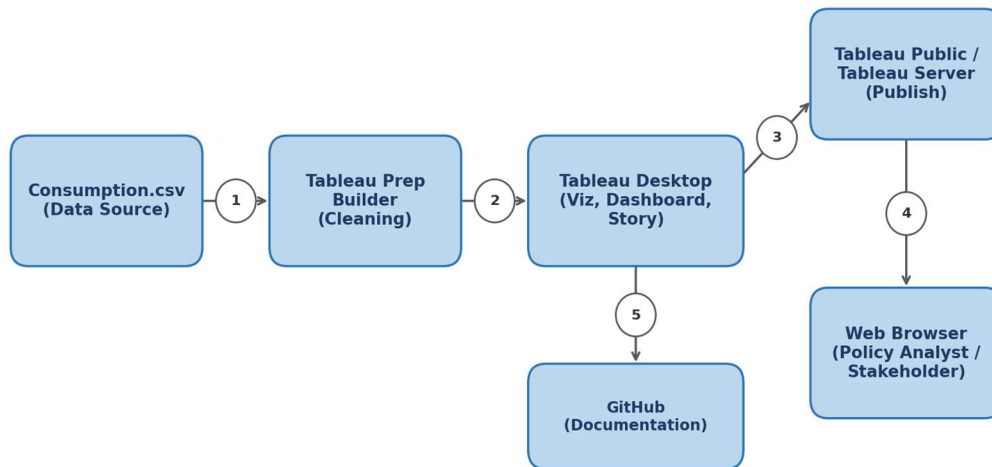


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	Data Source	Historical state-wise electricity consumption records (2019–2020)	CSV file (Consumption.csv)
2	Data Preparation Tool	Cleaning, deduplication, and field creation	Tableau Prep Builder
3	Visualization & Analysis Tool	Building charts, dashboards, and the story	Tableau Desktop
4	Calculated Fields	Custom metrics: lockdown-period flag, YoY growth, recovery rate	Tableau calculated fields (LOD expressions)
5	Geocoding / Mapping	Plotting state-wise consumption geographically	Built-in Tableau geocoding + CSV lat/long fields
6	Publishing Platform	Hosting the interactive dashboard and story	Tableau Public / Tableau Server
7	Web Embedding	Embedding the dashboard on a webpage	Tableau JavaScript API / HTML iframe embed
8	Version Control & Documentation	Tracking project files and documentation	GitHub
9	Collaboration & Tracking	Team coordination, sprint tracking	SkillWallet workspace / Kanban board

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Tools	Python/Pandas used for pre-cleaning checks alongside Tableau Prep	Python, Pandas (supporting tools)
2	Security	Data source access restricted to the project team; published dashboard visibility set appropriately (public/private)	Tableau permissions, restricted file access
3	Scalable Design	Extract-based data source lets new yearly data be appended without rebuilding the workbook	Tableau data extracts (.hyper)
4	Availability	Published dashboard accessible anytime via browser once hosted	Tableau Public / Tableau Server
5	Performance	Extracts used instead of a live connection for faster filtering and load times	Tableau extract (.hyper) optimization